

Chapter 2

Tuesday, January 30, 2024 8:33 AM

- Version Control
 - Manage changes in parallel code
 - Models
 - Pull request: Changed reviewed, code review style
 - Push: Commit to branch directly and pray, faster iteration
 - Recommended Git
 - Mercurial or SVN useful for other things
- Test Suite
 - Ensure correctness of code
 - Ensure no functions break
 - Parallelism: May affect numerical results due to reordering of operations etc.
 - Numdiff or ndiff can be used to test numerical differences between runs
- Memory
 - UB when using memory before it is initialized
 - Writing data to memory locations not owned or controlled results in failures
 - Check using Valgrind (also for MPI I think using special version?)
 - Integrates with GCC or Intel (extra steps)
- Portability
 - Make sure code can run on different systems / compilers
 - If one compiler works and another does not, then...
 - Compiler bug
 - Programming bug
- Profiling
 - Is memory, floating-point, or other factor such as IO the bottleneck?
 - Benchmark patterns of use
 - Tools can help identify areas to improve (python -m cProfile as example I use)
- Planning
 - Develop mini-apps
 - Smaller sections are easier to benchmark
 - Redeveloping efficient methods / code is cheaper
 - Plan data structure
 - Data structures must be well planned to ensure data moves correctly and efficiently
 - Shift to parallel
 - Algorithms must be possible to parallelize
 - May need to use different algorithms as things scale.
- Implementation
 - OpenMP for shared memory
 - MPI for distributed memory
 - Mixed OpenMP and MPI for high efficiency at human time cost
 - Assign tasks to specific people to allow them to work in manageable chunks.
- Commit
 - Start implementing code and apply testing before each commit
 - Committing often finds bugs earlier
 - Code will work until a single change, then commits cannot be reverted to fix problem!
 - Require high code quality standards for correctness (bugs) and readability (future work)
 - Test for portability as well if needed in the commit loop
 - This would be good use for CI on GitLab or GitHub